E-Commerce Sales & Customer Analytics Report

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Role: Data Analyst Project

Tools Used: SQL | Python | Power BI

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1. Executive Summary

This project focuses on analyzing e-commerce sales data to evaluate business performance, customer behavior, and profitability. The objective is to transform raw transactional data into meaningful insights using SQL, Python, and Power BI.

The analysis identifies key revenue drivers, loss-making products, and the impact of discounts on profit, enabling data-driven decision-making.

2. Problem Statement

Organizations often struggle to extract meaningful insights from large datasets. Without structured dashboards and analysis, it becomes difficult to identify top-performing products, high-value customers, and areas causing financial losses.

This project addresses these challenges by presenting data in a structured and visual format.

3. Objectives

- Analyze overall sales and profit performance
- Identify top customers and products
- Detect loss-making products
- Evaluate the impact of discounts on profitability
- Perform regional and category-level analysis

4. Dataset Overview

The dataset consists of transactional sales data including:

- **Order Information:** Order ID, Order Date, Ship Date
- **Customer Details:** Customer Name, Segment
- **Product Details:** Category, Sub-Category, Product Name
- **Sales Metrics:** Sales, Quantity, Discount, Profit
- **Geographical Data:** Country, Region, State, City

This dataset enables comprehensive analysis of business performance and customer behavior.

5. Tools & Technologies

- **SQL (MySQL):** Data extraction and querying
- **Python (Pandas, Matplotlib):** Data cleaning, transformation, and analysis
- **Power BI:** Interactive dashboard creation and visualization
- **Excel / CSV:** Data storage and preprocessing

6. Data Preparation

Data preparation was performed to ensure accuracy and consistency:

- Handled missing values and removed errors
- Corrected data types (date, numeric fields)
- Standardized date formats
- Created calculated measures using DAX
- Used Python for data validation and KPI calculations
- Used SQL for structured data querying

7. Key Performance Indicators (KPIs)

- **Total Sales:** 2M
- **Total Profit:** 286K
- **Total Orders:** 9,994
- **Average Order Value:** $229.9
- **Profit Margin:** 12.5%

These KPIs provide a high-level overview of business performance.

8. SQL Analysis

SQL was used to extract and analyze data efficiently.

Key queries performed:

- Total sales and profit calculation
- Top 10 customers by sales
- Top products by sales
- Identification of loss-making products
- Sales analysis by region and category

Figure 1: Overall Business Performance

total_sales	total_profit	total_orders
2297200.860299955	286397.0217000013	5009

Total sales and profit provide a high-level overview of business performance.

Figure 2: Sales and Profit by Region

Region	total_sales	total_profit
West	725457.8245000006	108418.44890000018
East	678781.2399999979	91522.78000000026
Central	501239.8908000005	39706.36249999998
South	391721.9050000003	46749.430300000065

The West region contributes the highest revenue compared to other regions.

Figure 3: Top 10 Customers by Sales

Customer Name	total_spent
Sean Miller	25043.05
Tamara Chand	19052.217999999997
Raymond Buch	15117.339
Tom Ashbrook	14595.62
Adrian Barton	14473.570999999998
Ken Lonsdale	14175.229
Sanjit Chand	14142.333999999999
Hunter Lopez	12873.297999999999
Sanjit Engle	12209.438000000002
Christopher Conant	12129.072

A small group of customers contributes a significant portion of total revenue.

Figure 4: Loss-Making Products

Product Name	total_loss
Cubify CubeX 3D Printer Double Head Print	-8879.9704
Lexmark MX611dhe Monochrome Laser Printer	-4589.973
Cubify CubeX 3D Printer Triple Head Print	-3839.9904
Chromcraft Bull-Nose Wood Oval Conference T...	-2876.1156
Bush Advantage Collection Racetrack Conferen...	-1934.3975999999998
GBC DocuBind P400 Electric Binding System	-1878.1662000000003
Cisco TelePresence System EX90 Videoconferen...	-1811.0784
Martin Yale Chadless Opener Electric Letter Ope...	-1299.1836
Balt Solid Wood Round Tables	-1201.0581
BoxOffice By Design Rectangular and Half-Moo...	-1148.4375
Riverside Furniture Oval Coffee Table, Oval En...	-1147.4
Epson TM-T88V Direct Thermal Printer - Monoch...	-1057.23

Certain products consistently generate negative profit and require attention.

Figure 5: Discount Impact on Profit

Discount	avg_profit
0	66.90029245518961
0.1	96.05507446808508
0.15	27.288298076923077
0.2	24.702572053595855
0.3	-45.679636123348004
0.32	-88.56065555555558
0.4	-111.92742912621364
0.45	-226.64646363636365
0.5	-310.703456060606
0.6	-43.077211594202915
0.7	-95.87405956937788
0.8	-101.79679733333329

Higher discounts are associated with reduced or negative profit.

9. Python Analysis

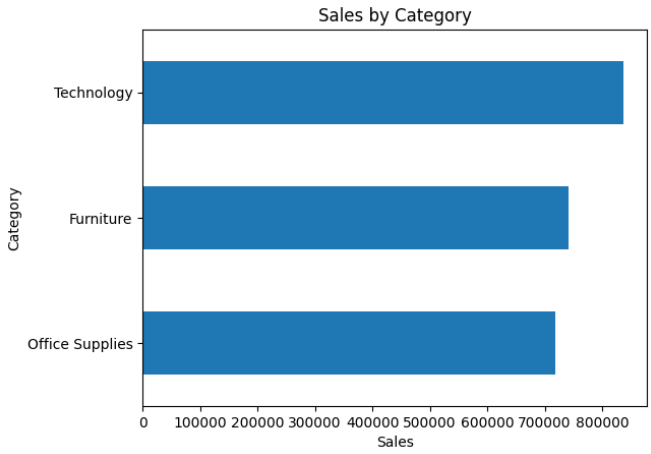
Python was used for exploratory data analysis and validation of insights.

Key tasks performed:

- Data cleaning and preprocessing
- KPI calculations
- Identification of top customers and products
- Detection of loss-making products
- Correlation analysis between discount and profit

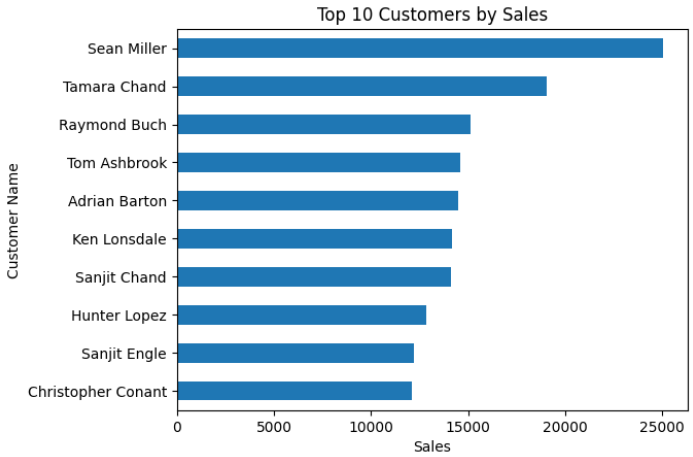
Visualizations created:

Figure 6: Sales by Category



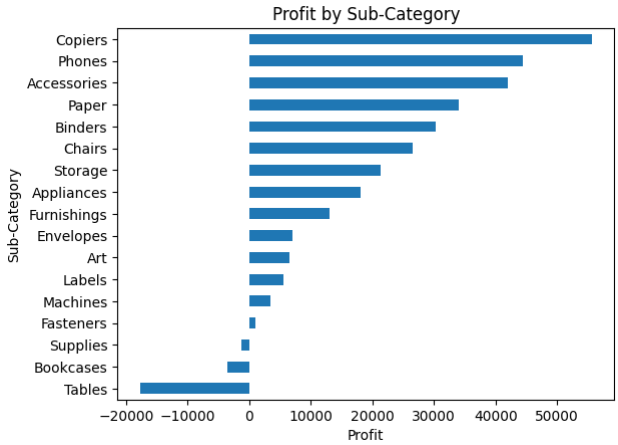
Technology category generates the highest sales among all categories.

Figure 7: Top 10 Customers by Sales



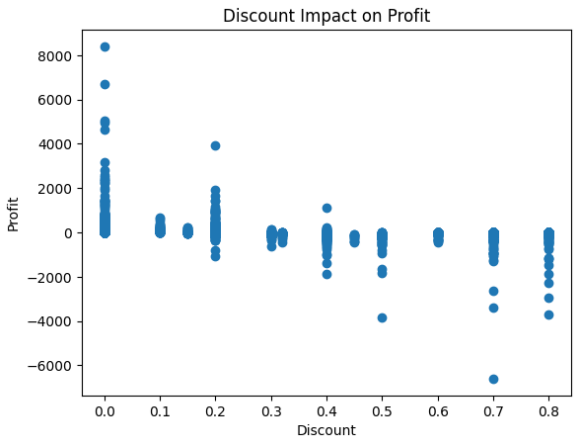
A small group of customers contributes significantly to total revenue.

Figure 8: Profit by Sub-Category



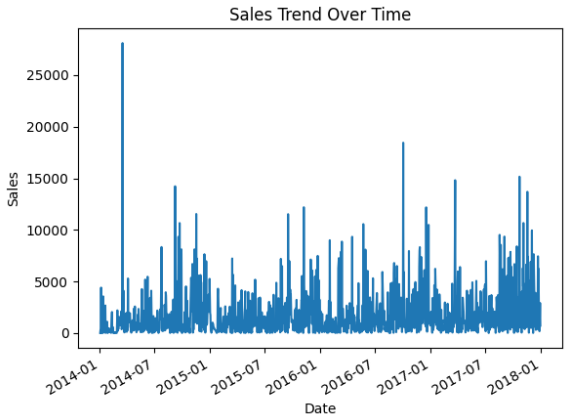
Some sub-categories generate high profit, while others underperform.

Figure 9: Discount Impact on Profit



Higher discounts are associated with lower or negative profit.

Figure 10: Sales Trend Over Time



Sales show a steady trend over time with gradual growth.

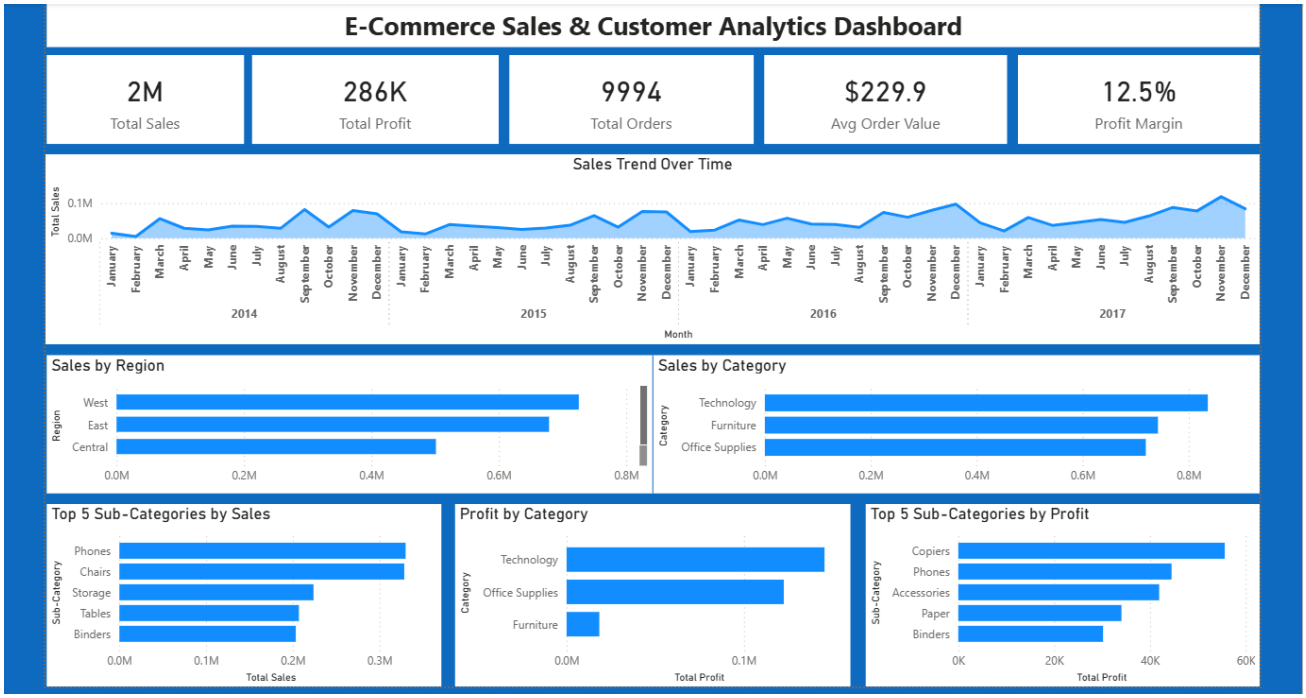
10. Power BI Dashboard

An interactive dashboard was developed using Power BI to visualize key business insights effectively.
The dashboard provides a comprehensive view of sales performance, customer behavior, and profitability trends.

Dashboard features include:

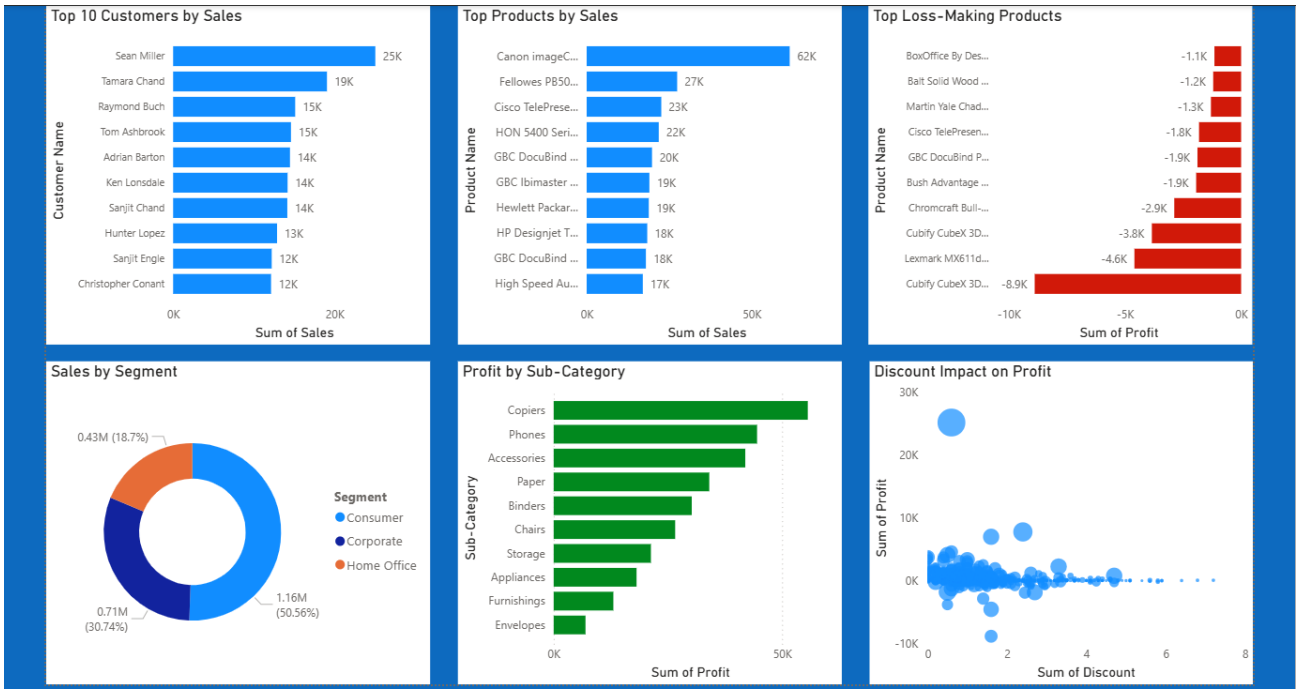
- KPI indicators for total sales, profit, orders, and profit margin
- Sales trend analysis over time
- Customer and product performance insights
- Profitability and loss analysis
- Discount impact on profit
- Regional and category-wise performance

Figure 11: Executive Overview Dashboard



The Executive Overview dashboard provides a high-level summary of business performance, including key metrics such as total sales, profit, orders, and profit margin.
It also highlights sales trends over time and regional performance.

Figure 12: Detailed Analysis Dashboard



The Detailed Analysis dashboard focuses on deeper insights, including top customers, top products, loss-making products, and discount impact on profitability.
It enables detailed exploration of factors affecting business performance.

11. Key Insights

- Technology category generates the highest sales and profit
- A small group of customers contributes a significant portion of revenue
- High discounts negatively impact profitability
- Certain products consistently generate losses
- Sales show a steady upward trend over time

12. Business Value

This dashboard helps businesses:

- Identify high-performing customers and products
- Reduce losses by addressing underperforming products
- Optimize discount strategies
- Improve decision-making using data-driven insights

13. Conclusion

This project demonstrates how data analysis and visualization can uncover valuable business insights. By integrating SQL, Python, and Power BI, a comprehensive analytical solution was developed.

The findings enable organizations to improve profitability, optimize strategies, and make informed decisions.

14. Recommendations

- Optimize discount strategies to protect profit margins
- Focus on high-value customers
- Improve or remove loss-making products
- Strengthen regional sales strategies

15. Future Scope

- Sales forecasting using machine learning
- Customer segmentation analysis
- Advanced predictive analytics